EXCEL ACADEMY

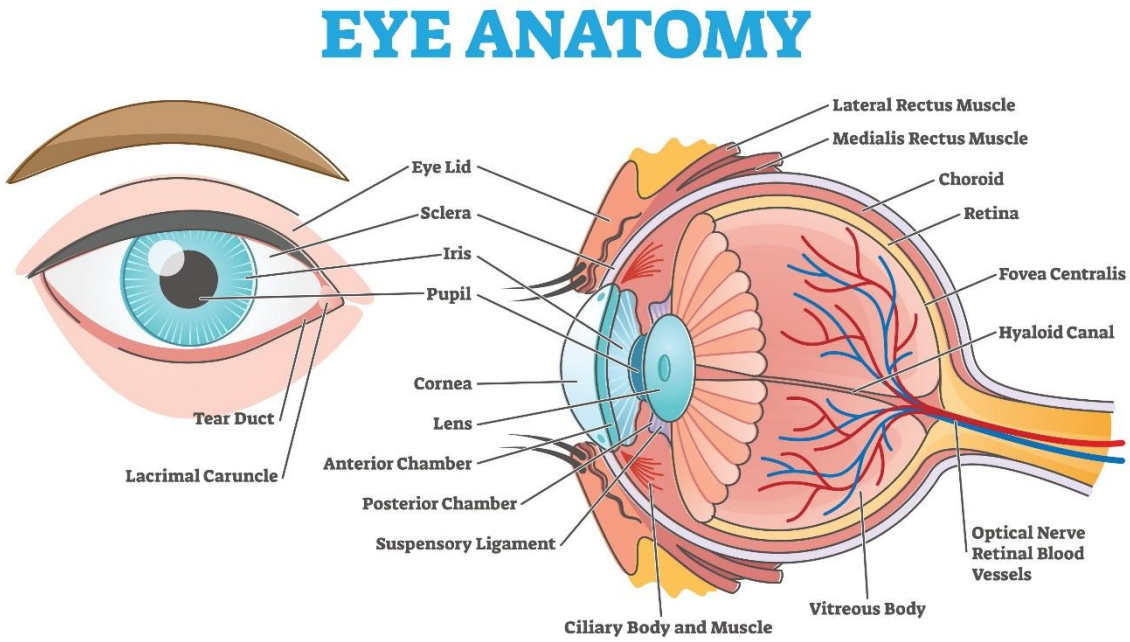
CLASS 10 NOTES

The Human Eye: Our Window to the World 👁️

The human eye is a remarkable and sensitive sense organ. It functions like a camera, enabling us to see the beautiful, colourful world around us. It forms an inverted, real image on a light-sensitive screen called the retina.

Structure of the Human Eye

The eyeball is approximately spherical with a diameter of about 2.3 cm. Let's look at its main parts.



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- **Cornea:** This is the transparent, dome-shaped outer layer at the front of the eye. Light enters the eye through the cornea. Its primary function is to **refract (bend) the light** as it enters.
- **Iris:** This is the coloured, muscular diaphragm behind the cornea. It controls the amount of light entering the eye by adjusting the size of the pupil. It's what gives people their eye colour (e.g., brown, blue, green).
- **Pupil:** This is the small black opening in the center of the iris. It **regulates the amount of light** entering the eye. In bright light, the iris contracts, making the pupil smaller. In dim light, the iris expands, making the pupil larger.
- **Crystalline Lens (Eye Lens):** Located behind the pupil, this is a transparent, biconvex lens made of a jelly-like material. Its function is to **fine-tune the focus** of light onto the retina. Its curvature can be changed by the ciliary muscles.
- **Ciliary Muscles:** These muscles are attached to the eye lens. They **change the focal length of the lens** by contracting and relaxing, which alters the lens's shape. This process is crucial for focusing on objects at different distances.

- **Retina:** This is the light-sensitive screen at the back of the eye, similar to the film in a camera. It contains a vast number of light-sensitive cells:
    - **Rods:** These cells are sensitive to the **intensity of light** (dim light vision).
    - **Cones:** These cells are sensitive to **colour** and function best in bright light.
- **Optic Nerve:** This nerve connects the retina to the brain. It transmits the electrical signals generated by the rod and cone cells to the brain, which then interprets these signals as images.
- **Aqueous Humour:** A clear, watery fluid that fills the space between the cornea and the lens. It maintains the pressure in the front part of the eye and provides nutrients.
- **Vitreous Humour:** A transparent, jelly-like substance that fills the space between the lens and the retina. It helps maintain the shape of the eyeball.

How We See: The Mechanism of Vision

1. Light rays from an object enter the eye through the cornea.
2. The cornea and the aqueous humour perform the initial, coarse refraction.
3. The iris adjusts the pupil size to control the amount of light entering.
4. The crystalline lens provides the final, fine adjustment of focal length to focus the light precisely on the retina.
5. A **real and inverted image** is formed on the retina.
6. The light-sensitive rod and cone cells on the retina get activated and generate electrical signals.
7. The optic nerve transmits these electrical signals to the brain.
8. The brain interprets these signals, processing them so that we perceive the object as **erect (upright)** and in its actual size and colour.

Power of Accommodation

The ability of the eye lens to adjust its focal length to see both nearby and distant objects clearly is called the **power of accommodation**.

- **For Distant Objects:** When looking at far-off objects, the ciliary muscles are **relaxed**. This allows the lens to become **thinner** (less convex), increasing its focal length and focusing the image on the retina.
- **For Nearby Objects:** When looking at nearby objects, the ciliary muscles **contract**. This increases the curvature of the lens, making it **thicker** (more convex). This shortens its focal length, allowing the image to be focused on the retina.

Key Points:

- **Far Point:** The maximum distance at which the eye can see objects clearly without any strain. For a normal eye, the far point is **infinity (∞)**.
- **Near Point (Least Distance of Distinct Vision):** The minimum distance at which an object can be seen clearly without strain. For a young adult with normal vision, the near point is about **25 cm**.

Defects of Vision and Their Correction 🤓

Sometimes, the eye cannot focus the image correctly on the retina, leading to blurred vision. These are called refractive defects of vision.

1. Myopia (Near-Sightedness)

A person with myopia can see **nearby objects clearly** but cannot see distant objects distinctly.

- **Cause:** This happens when the image of a distant object is formed **in front of the retina**. This can be due to:
    - Excessive curvature of the eye lens (lens is too thick).
    - Elongation of the eyeball.
- **Correction:** Myopia is corrected using a **concave lens** of suitable power. This lens diverges the incoming light rays before they enter the eye, effectively moving the final image back onto the retina.

2. Hypermetropia (Far-Sightedness)

A person with hypermetropia can see **distant objects clearly** but finds it difficult to see nearby objects.

- **Cause:** This happens when the image of a nearby object is formed **behind the retina**. This can be due to:
    - The focal length of the eye lens being too long (lens is too thin).
    - The eyeball being too short.
- **Correction:** Hypermetropia is corrected using a **convex lens** of suitable power. This lens converges the light rays before they enter the eye, helping the eye lens to focus the image onto the retina.

3. Presbyopia

This is the age-related difficulty in seeing nearby objects. It's essentially the hypermetropia of old age.

- **Cause:** It arises due to the gradual **weakening of the ciliary muscles** and the diminishing flexibility of the eye lens. The power of accommodation decreases, and the near point recedes.

- **Correction:** It is often corrected using **bifocal lenses**. The upper part of the lens is a concave lens (for distant vision), and the lower part is a convex lens (for near vision/reading).

4. Cataract

This is a medical condition in which the crystalline lens of the eye becomes milky and cloudy.

- **Cause:** It is typically an age-related condition.
- **Effect:** It leads to a partial or complete loss of vision.
- **Correction:** Vision can be restored through **cataract surgery**, where the cloudy lens is replaced with an artificial intraocular lens (IOL).

Interesting Facts about the Eye

- **Persistence of Vision:** The image of an object continues to persist on the retina for about 1/16th of a second even after the object is removed. This phenomenon is responsible for the illusion of motion in movies and animation.
- **Colour Perception:** The cone cells are responsible for colour vision. We have cones that are sensitive to red, green, and blue light. The brain perceives other colours based on the combination and degree of stimulation of these cones.
- **Why we have two eyes:** Having two eyes gives us a wider field of view and enables stereoscopic vision (stereopsis), which gives us a three-dimensional perspective and helps in depth perception.

Physics Principles of the Human Eye 🔬

The human eye is a fascinating optical instrument that operates on several fundamental principles of physics.

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1. Refraction and Image Formation

The primary principle behind vision is the **refraction of light**. The eye functions as a converging lens system to form an image on a light-sensitive screen.

- **Lens System:** The eye's cornea and the crystalline lens together act as a **convex lens**.
- **Refraction Points:** Light is refracted first and most significantly by the cornea, and then its focus is fine-tuned by the crystalline lens.
- **Image Characteristics:** This lens system forms a **real, inverted, and diminished** image of the object on the retina. The brain then processes this inverted image to perceive it as upright.

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2. Power of Accommodation and Focal Length

This is the eye's ability to see both distant and near objects clearly by adjusting its focal length.

- **Variable Focal Length:** Unlike a camera with a fixed lens that moves, the eye changes the **shape and focal length** of its crystalline lens.
- **Ciliary Muscles:** These muscles control the lens. When they relax, the lens becomes thinner, its focal length **increases**, and its converging power **decreases** (for seeing distant objects). When they contract, the lens becomes thicker, its focal length **decreases**, and its power **increases** (for seeing near objects).
- **Power Formula:** The power of a lens is given by $P=1/f$, where f is the focal length in meters. As the focal length changes, so does the power of the eye lens.

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3. Aperture Control (Pupil and Iris)

The eye can control the amount of light entering it, just like the aperture of a camera.

- **Iris as Diaphragm:** The iris acts as a muscular diaphragm.
- **Pupil as Aperture:** The pupil is the aperture or opening. In bright light, the iris contracts, making the pupil smaller to limit light entry. In dim light, the iris expands, widening the pupil to allow more light in.

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4. Optical Defects and Correction

Defects of vision are essentially problems with the eye's focusing system, which are corrected using the principles of optics.

- **Myopia (Near-sightedness):** The eye's converging power is too high, or the eyeball is too long. The image forms *in front* of the retina. This is corrected by placing a **concave (diverging) lens** in front of the eye.
- **Hypermetropia (Far-sightedness):** The eye's converging power is too low, or the eyeball is too short. The image forms *behind* the retina. This is corrected by using a **convex (converging) lens**.

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5. Wavelength and Colour Perception

Our ability to see colour is based on the physics of light waves.

- **Visible Spectrum:** The retina's cone cells are sensitive to different **wavelengths** of light within the visible spectrum.
- **Photoreceptors:** There are three types of cones, primarily sensitive to the wavelengths corresponding to **red, green, and blue light**. The brain interprets the different combinations of signals from these cones to perceive the entire spectrum of colours.